

TYUMEN ROSCHINO, Russian Federation

WMO: 232204

Lat: 57.190N

Lon: 65.324E

Elev: 378

StdP: 14.50

Time Zone: 5.00 (E05)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.6	-20.6	-31.2	1.0	-25.0	-26.0	1.3	-20.0	20.3	17.5	18.7	15.6	5.7	230	0.681

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	85.9	67.7	82.6	66.1	79.2	64.6	70.2	80.7	68.5	78.6	67.0	76.6	9.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.5	98.8	74.6	64.7	92.8	72.5	63.0	87.2	70.9	34.4	80.7	33.1	78.6	31.8	76.3	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	18.4	16.5	DB	-30.2	91.0	6.3	2.7	-34.8	92.9	-38.5	94.5	-42.0	96.0	-46.6	97.9
			WB	-30.4	72.5	5.9	1.7	-34.7	73.7	-38.2	74.6	-41.5	75.6	-45.8	76.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.6	2.3	7.6	23.6	40.7	53.0	62.8	65.9	61.8	51.1	37.7	21.9	9.4	(6)
(7)		DBStd	24.29	13.45	12.23	8.74	9.55	9.58	8.22	6.92	7.53	7.75	9.29	12.97	13.92	(7)
(8)		HDD50	6455	1479	1188	818	305	81	8	0	3	78	393	842	1260	(8)
(9)		HDD65	10605	1944	1608	1283	730	387	136	77	155	421	847	1292	1725	(9)
(10)		CDD50	1575	0	0	0	25	174	392	493	369	111	11	0	0	(10)
(11)		CDD65	250	0	0	0	0	15	70	105	56	4	0	0	0	(11)
(12)		CDH74	2351	0	0	0	8	207	700	917	485	34	0	0	0	(12)
(13)		CDH80	609	0	0	0	0	38	202	231	135	3	0	0	0	(13)
(14)	Wind	WSAvg	9.0	8.7	8.6	10.0	10.1	9.6	8.8	8.2	8.2	8.6	9.4	9.4	8.9	(14)
(15)	Precipitation	PrecAvg	19.1	0.9	0.6	0.9	0.9	1.7	2.3	3.5	2.4	1.9	1.5	1.4	1.1	(15)
(16)		PrecMax	26.5	1.9	1.3	2.2	3.1	4.1	5.0	6.1	4.5	4.4	3.7	2.8	2.1	(16)
(17)		PrecMin	13.9	0.1	0.1	0.1	0.0	0.2	0.7	0.6	0.6	0.4	0.1	0.1	0.2	(17)
(18)		PrecStd	3.2	0.5	0.3	0.5	0.8	1.0	1.1	1.5	0.9	1.0	0.9	0.6	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	37.6	47.8	73.5	84.3	89.7	89.3	88.2	78.6	66.1	48.2	37.3	(19)
(20)		MCWB	33.4	34.0	39.7	56.0	61.1	66.7	69.1	67.9	60.4	53.2	43.0	35.3	(20)	
(21)		2%	DB	29.9	33.8	42.5	66.0	79.2	85.7	85.7	84.0	72.0	59.4	42.9	34.0	(21)
(22)		MCWB	29.0	31.5	36.1	51.8	58.7	66.6	69.2	67.5	59.2	49.7	40.9	32.7	(22)	
(23)		5%	DB	25.1	28.8	38.9	60.4	75.3	82.1	82.4	79.1	67.8	55.1	39.6	31.6	(23)
(24)		MCWB	24.5	26.7	34.3	48.2	57.3	65.4	67.5	66.3	57.7	48.0	38.3	30.9	(24)	
(25)		10%	DB	21.2	24.8	35.9	55.3	70.2	77.4	79.1	75.3	63.0	50.2	37.0	27.0	(25)
(26)		MCWB	20.3	23.1	32.5	45.7	55.0	64.2	65.7	64.5	55.1	45.8	35.7	26.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.0	34.6	40.1	57.2	65.3	71.7	72.7	70.9	63.7	54.2	43.7	35.1	(27)
(28)		MDCB	33.9	36.7	46.0	66.9	76.7	81.4	82.8	81.6	74.0	63.3	46.1	36.2	(28)	
(29)		2%	WB	28.4	31.3	36.9	52.9	61.9	69.0	70.7	69.1	61.0	51.2	41.0	32.8	(29)
(30)		MDCB	29.2	33.8	40.7	64.1	73.6	79.9	80.9	79.9	69.2	58.0	42.3	33.8	(30)	
(31)		5%	WB	24.6	26.8	34.8	49.6	59.1	67.1	69.2	67.3	58.6	48.5	38.5	30.3	(31)
(32)		MDCB	25.3	28.7	37.6	58.2	71.4	77.8	79.2	77.0	65.5	53.4	39.6	31.1	(32)	
(33)		10%	WB	20.2	22.9	32.9	46.3	56.3	65.3	67.5	65.4	56.1	46.2	35.2	26.2	(33)
(34)		MDCB	20.9	24.7	35.7	53.4	68.2	75.2	76.7	73.9	62.4	50.4	36.3	27.2	(34)	
(35)	Mean Daily Temperature Range	MDBR	12.4	16.0	15.9	17.0	20.8	19.2	17.9	17.5	15.7	12.5	10.3	11.5	(35)	
(36)		5% DB	MCDBR	11.6	14.2	16.3	23.7	26.2	24.5	22.9	23.0	21.0	19.2	10.8	9.5	(36)
(37)		MCWBR	11.2	12.6	11.9	12.9	12.3	10.6	9.7	10.2	10.9	11.6	9.2	9.1	(37)	
(38)		5% WB	MCDBR	11.3	14.3	14.5	21.7	23.4	21.5	20.4	21.0	19.1	17.6	10.6	9.7	(38)
(39)		MCWBR	11.1	13.1	11.5	12.8	13.3	11.0	9.8	10.4	11.2	11.7	9.5	9.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.279	0.309	0.366	0.407	0.404	0.403	0.419	0.427	0.380	0.346	0.303	0.278	(40)	
(41)		taud	2.040	2.038	1.957	2.098	2.240	2.314	2.272	2.218	2.314	2.271	2.132	1.998	(41)	
(42)		Ebn at Noon	176	224	240	252	264	266	258	245	240	213	166	136	(42)	
(43)		Edh at Noon	19	30	42	43	40	37	38	38	30	23	17	15	(43)	
(44)	All-Sky Solar Radiation	RadAvg	223	535	1012	1409	1651	1807	1736	1333	838	426	206	134	(44)	
(45)		RadStd	16	37	56	111	140	139	154	129	85	68	29	17	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page